20CYS402 Distridbuted Systems and Cloud Computing

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CB.SC.U4CYS23013

Docker Build

```
PS C:\GT\Distributed systems\cb.sc.u4cys23013_MIDTERM> docker build -t mt-webauth .
[+] Building 2.3s (12/12) FINISHED                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                0.0s
 => => transferring dockerfile: 566B                                                0.0s
 => [internal] load metadata for docker.io/library/ubuntu:22.04                     1.9s
 => [auth] library/ubuntu:pull token for registry-1.docker.io                       0.0s
 => [internal] load .dockerignore                                                   0.0s
 => => transferring context: 2B                                                     0.0s
 => [1/6] FROM docker.io/library/ubuntu:22.04@sha256:3b06811b2afd352be909dd088a004166d  0.0s
 => => resolve docker.io/library/ubuntu:22.04@sha256:3b06811b2afd352be909dd088a004166d  0.0s
 => [internal] load build context                                                   0.0s
 => => transferring context: 120B                                                   0.0s
 => CACHED [2/6] WORKDIR /midterm-exam                                              0.0s
 => CACHED [3/6] RUN apt-get update &&     apt-get install -y python3 python3-pip mysq  0.0s
 => CACHED [4/6] COPY database/init.sql .                                           0.0s
 => CACHED [5/6] COPY web/app.py .                                                  0.0s
 => CACHED [6/6] RUN service mysql start &&     mysql < init.sql                    0.0s
 => exporting to image                                                              0.1s
 => => exporting layers                                                             0.0s
 => => exporting manifest sha256:083fd7d6b2868d5423af5b30d0785c4c8081470b898b6d221f1d3  0.0s
 => => exporting config sha256:21d094dd4b21be85c39b067e4a777440b2bf3abcb066ccf7656abaa  0.0s
 => => exporting attestation manifest sha256:68fb47af57cd801ad700a82679e66d211c92fe3ea  0.0s
 => => exporting manifest list sha256:66abeb9a39caa08dac29ccca165963f48d9e8ede658d991c  0.0s
 => => naming to docker.io/library/mt-webauth:latest                                0.0s
 => => unpacking to docker.io/library/mt-webauth:latest                             0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/5afruc4grmx0cmme07icztj29
PS C:\GT\Distributed systems\cb.sc.u4cys23013_MIDTERM>
```

Dockerfile

```
FROM ubuntu:22.04

ENV ROLLNUMBER="CB.SC.U4CYS23013"
ENV NAME="Gautham"

LABEL EMAIL="cb.sc.u4cys23013@cb.students.amrita.edu"
LABEL "PRODUCTION VERSION"="1.0"

WORKDIR /midterm-exam

RUN apt-get update && \
    apt-get install -y python3 python3-pip mysql-server && \
    pip3 install flask mysql-connector-python && \
    apt-get clean

COPY database/init.sql .
COPY web/app.py .
```

```
RUN service mysql start && \
    mysql < init.sql

EXPOSE 5000

CMD ["sh", "-c", "service mysql start && python3 app.py"]
```

Docker-compose.yml

```
services:
  webauth:
    build: .
    image: mt-webauth
    ports:
      - "5000:5000"
    volumes:
      - ./web:/midterm-exam
```

App.py

```
from flask import Flask, request
import mysql.connector

app = Flask(__name__)

@app.route("/", methods=["GET", "POST"])
def home():

    message = ""

    if request.method == "POST":

        username = request.form["username"]
        password = request.form["password"]

        db = mysql.connector.connect(
            host="localhost",
            user="midterm_user",
            password="midterm_pass",
            database="midterm_db"
        )

        cursor = db.cursor()

        cursor.execute(
            "SELECT * FROM users WHERE username=%s AND password=%s",
```

```
            (username, password)
        )

        result = cursor.fetchone()

        if result:
            message = "Login successful!"
        else:
            message = "Invalid username or password!"

        cursor.close()
        db.close()

    return """
    <h1>Midterm Web Authentication</h1>

    <form method="POST">

    Username:
    <input type="text" name="username">
    <br><br>

    Password:
    <input type="password" name="password">
    <br><br>

    <input type="submit" value="Login">

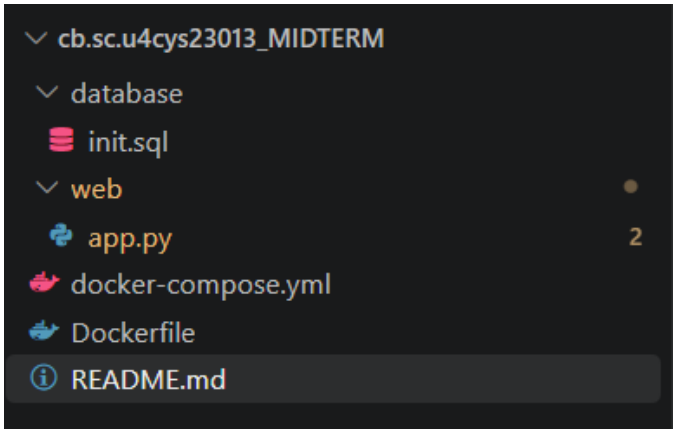
    </form>

    <h3>{}</h3>

    """.format(message)


app.run(host="0.0.0.0", port=5000)
```

Structure



Database

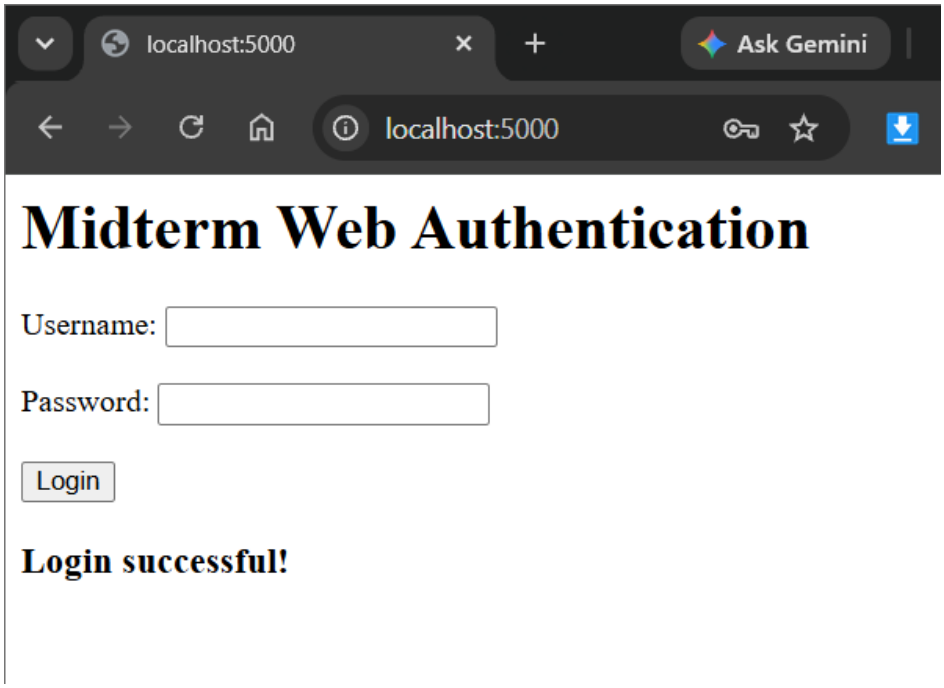
```
Database changed
mysql> SHOW TABLES;
+----------------------+
| Tables_in_midterm_db |
+----------------------+
| users                |
+----------------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM users;
+----+----------+----------+
| id | username | password |
+----+----------+----------+
|  1 | gt       | dog@123  |
+----+----------+----------+
1 row in set (0.00 sec)
```

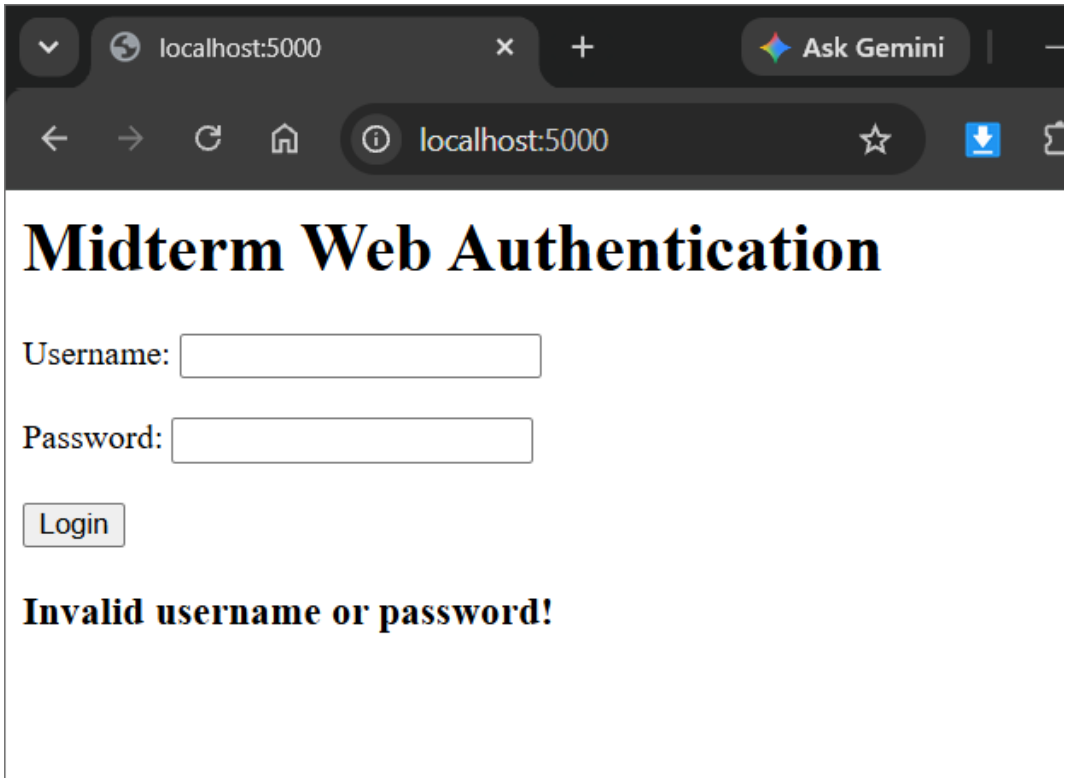
Declared ENV

```
root@698ce8816cc2:/midterm-exam# echo $ROLLNUMBER
CB.SC.U4CYS23013
root@698ce8816cc2:/midterm-exam# echo $NAME
Gautham
```

Login check :Correct Password



Wrong password



Docker compose

```
PS C:\GT\Distributed systems\cb.sc.u4cys23013_MIDTERM> docker compose up
Attaching to webauth-1
webauth-1  |  * Starting MySQL database server mysqld
webauth-1  | su: warning: cannot change directory to /nonexistent: No such file or directory
webauth-1  |    ...done.
webauth-1  |  * Serving Flask app 'app'
webauth-1  |  * Debug mode: off
webauth-1  | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI serv
er instead.
webauth-1  |  * Running on all addresses (0.0.0.0)
webauth-1  |  * Running on http://127.0.0.1:5000
webauth-1  |  * Running on http://172.20.0.2:5000
webauth-1  | Press CTRL+C to quit
webauth-1  | 172.20.0.1 - - [14/Aug/2026 10:21:44] "GET / HTTP/1.1" 200 -
webauth-1  | 172.20.0.1 - - [14/Aug/2026 10:21:44] "GET /favicon.ico HTTP/1.1" 404 -
webauth-1  | 172.20.0.1 - - [14/Aug/2026 10:21:44] "GET / HTTP/1.1" 200 -
webauth-1  | 172.20.0.1 - - [14/Aug/2026 10:30:08] "POST / HTTP/1.1" 200 -
webauth-1  | 172.20.0.1 - - [14/Aug/2026 10:30:08] "GET / HTTP/1.1" 200 -
webauth-1  | 172.20.0.1 - - [14/Aug/2026 10:30:12] "POST / HTTP/1.1" 200 -
webauth-1  | 172.20.0.1 - - [14/Aug/2026 10:30:12] "GET / HTTP/1.1" 200 -
```

Running container

```
PS C:\Users\GOWTHAM> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                       NAMES
698ce8816cc2   cb9316a3cfc0   "sh -c 'service mysq…"   45 minutes ago   Up 28 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   cbscu4cys23013_midterm
-webauth-1
```

Pushed

```
PS C:\GT\Distributed systems\cb.sc.u4cys23013_MIDTERM> docker push gt57/mt-webauth:latest
The push refers to repository [docker.io/gt57/mt-webauth]
c92e563be125: Pushed
81e4033eba19: Pushed
94c3a8d55e5f: Pushed
8a8e16af54e4: Pushed
de5bf8f98749: Pushed
b671d28f13ed: Pushed
39a945af8df2: Pushed
latest: digest: sha256:cb9316a3cfc008bc30f23de4db34df4c2797995da6424c8df97ec98cdcbb9a1a size: 856
```